

Year Two Lecture Notes

BSc Computer Science - Durham University

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Part I

Artificial Intelligence

1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

This module is assessed through three pieces of summative coursework, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	10/10/2026	16/12/2026	22/01/2026	Dr Yang Long	50%
AI Search	12/01/2027	19/02/2027	19/03/2026	Prof. Iain Stewart	25%
Bias in AI	06/02/2027	26/03/2027	07/05/2027	Dr. Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	n/a
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

This section is taught by Dr Yang Long. The recommended reading for this topic is:

- Mathematics for Machine Learning (Deisenroth, M.P., Faisal, A.A., and Ong, C.S., 2020).
- Mastering Machine Learning with scikit-learn (Hackeling, G., 2017).

2.1 Introduction to Machine Learning

2.2 Machine Learning Workflow and Coding Environment

2.3 Data Understanding

2.4 Data Preparation Workflow

3 Artificial Intelligence Search

This section is taught by Professor Iain Stewart. There is no recommended reading for this topic however there may be an additional video to watch after lectures to enable more content to be covered.

Watch the FAQs video.

3.1 Coding Real-World Problems for Search Tree Expansion

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.2 Uninformed Search: BFS, DFS, and Iterative Deepening

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.3 General Heuristic Search: Greedy Best-First Search and A* Search

3.4 A* Search and Deriving Iterative Deepening A* Search

3.5 Heuristic Functions and Their Development

3.6 Local Search: Hill-Climbing, Simulated Annealing, and Genetic Algorithms

3.7 Swarm Behaviour, How Ants Forage, and a Simple S-ACO algorithm

3.8 Applying Different Ant Systems to the TSP

3.9 General Versions of Particle Swarm Optimisation and Differential Evolution

3.10 Applying Particle Swarm Optimization to the TSP

4 Ethics and Bias in AI

This section is taught by Dr Shauna Concannon. The recommended reading is handled on a weekly basis for this topic.

4.1 Introduction to AI Ethics and Related Concepts

4.2 Societal Impacts of AI

4.3 Defining Bias and Fairness in ML

4.4 How to Build Better Systems

4.5 Value Sensitive Design

4.6 Evaluations and Audits

4.7 Bias Mitigation Strategies

4.8 Transparency and Accountability for Trustworthy Systems

4.9 Explainable AI

4.10 Future Trends and Ethical Frontiers

Part II

Data Science

5 Module Introduction

Data Science has the module code COMP2271 and is led by Professor Thomas Erlebach.

5.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 3. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Data Cleaning & Analytics	Week 11	n/a	13/02/2027	Dr Igor Razgon	25%
Image Processing	16/01/2027	26/02/2027	26/03/2027	Dr George Kouliris	25%
Probability and Graphics	May Exam	n/a	n/a	Razgon & Kouliris	50%

Table 3: Assessments for Data Science.

5.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 4.

Day	Time	Class Type	Location	Lecturer
Monday	12:00 - 13:00	Lecture	W103	Dr Igor Razgon
Friday	10:00 - 11:00	Lecture	W103	Dr Igor Razgon
Friday	16:00 - 18:00	Computer Class	CC-0007	n/a

Table 4: Michaelmas Term Timetable for Data Science.

6 Probability and Statistics

This section is taught by Dr Igor Razgon. The recommended reading for this topic is:

- Probability and Statistics (Degroot, M.H. and Schervish, M.J., 2012).

6.1 Probability and Statistics: Basic Concepts

6.1.1 Counting Theory

Definition 6.1.1.1: The product rule states that, if a procedure can be broken down into a sequence of two tasks with n_1 ways to do the first task and n_2 ways to do the second task, then there are $n_1 n_2$ ways to complete the entire procedure.

Example 6.1.1.1: John wants to fly to Canada from Germany with a connecting flight through the UK. There are 3 flights from Germany to the UK, and 2 flights from the UK to Canada. How many possible flight combinations are there?

Solution: $3 \times 2 = 6$.

□

Definition 6.1.1.2: The sum rule states that, if you consider two independent sub-tasks (we only have to do one), if the first can be done in n_1 ways, and the second can be done in n_2 ways, then the total number of ways to complete the task is $n_1 + n_2$.

Example 6.1.1.2: Mary wants to choose a student project from three lists: lecturer 1 offers 5 projects, lecturer 2 offers 2 projects, and lecturer 3 offers 10 projects. How many possible options for her?

Solution: $5 + 2 + 10 = 17$.

□

Definition 6.1.1.3: The inclusion-exclusion principle states that, if two tasks can be done at the same time (unlike the sum rule in which they are independent), then there is a possibility of overcounting since some ways may be counted twice. To counter this, we add up the number of ways to do each task, then subtract the number of ways to do both tasks.

$$|A \cup B| = (|A| + |B|) - (|A \cap B|) \quad (1)$$

$$|A \cup B \cup C| = (|A| + |B| + |C|) - (|A \cap B| + |A \cap C| + |B \cap C|) + (|A \cap B \cap C|) \quad (2)$$

Definition 6.1.1.4: An r-permutation of a set of r distinct objects is one ordered arrangement. We denote the number of r-permutations of a set with n elements in as $P(n, r)$.

$$P(n, r) = n \times (n - 1) \times (n - 2) \times \dots \times (n - r + 1) = \frac{n!}{(n - r)!} \quad (3)$$

Suppose that a set consists of n objects, of which n_1 are of one type (indistinguishable from one another), n_2 are of a second type, ..., and n_k are of a k_{th} type where $n = n_1 + n_2 + \dots + n_k$. The number of permutations of the objects can be calculated with (4).

$$P(n; n_1, n_2, \dots, n_k) = \frac{n!}{n_1! n_2! \dots n_k!} \quad (4)$$

Definition 6.1.1.5: A r-combination is a counting of distinct objects in an unordered arrangement of length r . The number of r combinations of a set with n distinct elements is given by (5).

$$C(n, r) = \binom{n}{r} = \frac{n!}{(n - r)! \times r!} \quad (5)$$

Example 6.1.1.3: Assume elements A_1, A_2, A_3, A_4 are in a set. We want to choose 2 elements from them.

Solution:

- If the order of elements is important: $P(4, 2) = \frac{4!}{(4-2)!} = 12$.
- If the order of elements is not important: $C(4, 2) = \frac{4!}{(4-2)!2!} = 6$.

□

6.1.2 Probability Theory

Definition 6.1.2.1: Probability is the formalisation of uncertain events. It enables one to infer probabilities of interest based on assumptions and observations.

Definition 6.1.2.2: Statistics is the practice of the science of collecting and analysing data. It deals with experimental data and extracting knowledge from it.

- 6.2 Dependency and Bayes Theorem
- 6.3 Discrete Probability Distributions
- 6.4 Continuous Probability Distributions
- 6.5 Gaussian Distribution and its Characteristics
- 6.6 Correlations
- 6.7 Central Limit Theorem and Sampling
- 6.8 Hypothesis Testing
- 6.9 Data Types and Visualisation
- 6.10 Basic Data Analysis with Python
- 7 Data Cleaning and Analytics

This section is taught by Dr Igor Razgon. [Check for recommended reading.](#)

8 Computer Graphics

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

9 Image Processing

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

Part III

Networks and Systems*

10 Module Introduction

Networks and Systems has the module code COMP2211 and is led by Dr Wanqing Tu.

10.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 5. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Cyber Security	10/10/2026	06/11/2026	04/12/2026	Dr Carlton Shepherd	25%
Networks	31/10/2026	08/01/2027	05/02/2027	Dr Wanqing Tu	25%
Dist. Sys. & Databases	May Exam	n/a	n/a	Dr Vasantam	50%

Table 5: Assessments for Networks and Systems.

10.2 Timetable

This module is taught through one two-hour lecture and one two-hour computer class each week as seen below in Table 6.

Day	Time	Class Type	Location	Lecturer
Thursday	09:00 - 11:00	Lecture	CLC202	Dr Carlton Shepherd
Thursday	16:00 - 18:00	Computer Class	CC-0007	n/a

Table 6: Michaelmas Term Timetable for Networks and Systems.

11 Cyber Security

This section is taught by Dr Carlton Shepherd. **Check for recommended reading.**

11.1 Introduction, Threat Modelling, and Applied Cryptography

12 Networks

This section is taught by Dr Wanqing Tu. **Check for recommended reading.**

13 Databases

This section is taught by Dr Vasantam. The recommended reading for this topic is:

- Connolly, T.M. and Begg, C.E. (2015). Database Systems: A practical approach to design, implementation, and management. 6th edn. Harlow, Essex, England: Pearson Education Limited.

- 13.1 Enhanced Entity-Relationship Model**
- 13.2 Semistructured Databases (XML)**
- 13.3 XML Data Manipulation (XPath and XQuery)**
- 13.4 Transactions**
- 13.5 Concurrency Control**
- 13.6 Distributed Transactions**
- 13.7 Distributed Concurrency Control**

14 Distributed Systems

This section is taught by Dr Vasantam. The recommended reading for this topic is:

- Van Steen, M. and Tanenbaum, A.S. (2025). Distributed Systems. 4th edn. [online] Self-Published. Available at: <https://www.distributed-systems.net/index.php/books/ds4/> [Accessed 5 Oct. 2026].

- 14.1 Introduction to Distributed Systems**
- 14.2 Middleware and RMI Technologies**
- 14.3 Message-Oriented Middleware and Web Services**
- 14.4 Replication Models**
- 14.5 Concurrent Events and Consistency Control**
- 14.6 Fault Tolerance**
- 14.7 Load Distribution**

Part IV

Programming Paradigms

15 Module Introduction

Programming Paradigms has the module code COMP2221 and is led by Dr Stuart James.

15.1 Assessment

This module is assessed through one piece of summative coursework and two final exams, these can be seen in Table 7. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Systems Programming	31/10/2026	11/12/2026	08/01/2027	Dr Stuart James	50%
Object Oriented & Functional	May Exam	n/a	n/a	Ivkovic & Moses	50%

Table 7: Assessments for Programming Paradigms.

15.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 8.

Day	Time	Class Type	Location	Lecturer
Monday	11:00 - 12:00	Lecture	CLC202	Dr Stuart James
Monday	16:00 - 18:00	Computer Class	CB-LG001	n/a
Friday	14:00 - 15:00	Lecture	CLC202	Dr Stuart James

Table 8: Michaelmas Term Timetable for Programming Paradigms.

16 Systems Programming

This section is taught by Dr Stuart James. There is no official reading list for this module.

Definition 16.0.0.1: Systems programming involves the development of individual pieces of software that allow the entire system to function as a single unit.

16.1 Version Control and UNIX

16.2 The Basics of C

16.3 Control Flow and Functions

16.4 Data Types and Structures

16.5 Advanced Bash Scripting

16.6 Pointers

16.7 Memory Management

16.8 Debugging

16.9 Scope and Recursion

16.10 Libraries

16.11 stdlib

16.12 C++

17 Functional Programming

This section is taught by Dr Billy Moses Jr.. [Check for recommended reading.](#)

18 Object Oriented Programming

This section is taught by Dr Mladen Ivkovic. [Check for recommended reading.](#)

18.1 Java Fundamentals

18.2 OOP with Classes and Objects

18.3 Encapsulation

18.4 Inheritance

18.5 Abstraction with Abstract Classes

18.6 Abstraction with Interfaces

18.7 Polymorphism

18.8 Java Collections

18.9 Best Practises

Part V

Software Engineering

19 Module Introduction

Software Engineering has the module code COMP2281 and is led by Dr Craig Stewart.

19.1 Assessment

This module is assessed through a group project, the deadlines for this can be seen in Table 9. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Peer Evaluation 1	17/10/2026	20/11/2026	22/01/2027	Dr Craig Stewart	1.67%
Requirement Documentation	17/10/2026	20/11/2026	22/01/2027	Dr Craig Stewart	15%
Peer Evaluation 2	17/10/2026	12/02/2027	07/05/2027	Dr Craig Stewart	1.67%
Technical Report	17/10/2026	12/03/2027	07/05/2027	Dr Craig Stewart	30%
Product Presentation	17/10/2026	20/03/2027	07/05/2027	Dr Craig Stewart	15%
Peer Evaluation 3	17/10/2026	24/03/2027	07/05/2027	Dr Craig Stewart	1.67%
Reflective Experience Report	17/10/2026	30/04/2027	n/a	Dr Craig Stewart	20%
Project Handover	17/10/2026	30/04/2027	n/a	Dr Craig Stewart	10%
Git Commit	17/10/2026	30/04/2027	n/a	Dr Craig Stewart	5%

Table 9: Assessments for Software Engineering.

19.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 10.

Day	Time	Class Type	Location	Lecturer
Tuesday	09:00 - 10:00	Lecture	TLC033	Dr Nelly Bencomo
Wednesday	11:00 - 12:00	Lecture	TLC033	Dr Nelly Bencomo
Thursday	11:00 - 13:00	Practical	CC-0007	Dr Gagangeet Aujla

Table 10: Michaelmas Term Timetable for Software Engineering.

Add lectures into relevant sections when you do them with the lecturer linked. Check for recommended reading.

19.3 Introduction to the Module

20 Project Management

21 Software Engineering

22 Legal, Social, Ethical, and Professional

23 Group Project

Part VI

Theory of Computation

24 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

24.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 11.

Title	Opening	Due	Feedback	Professor	Weight
Benchtest 1	Week 11	n/a	09/02/2027	Dr Dantchev and Dr Mertzios	17%
Benchtest 2	Week 20	n/a	07/05/2027	Dr Dantchev and Dr Mertzios	17%
Final Exam	May Exam	n/a	n/a	Dr Barnaby Martin	66%

Table 11: Assessments for Theory of Computation.

24.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 12.

Day	Time	Class Type	Location	Lecturer
Tuesday	10:00 - 11:00	Lecture	CLC202	Dr Barnaby Martin
Wednesday	09:00 - 11:00	Problem Class	TLC116	n/a
Thursday	14:00 - 15:00	Lecture	CLC202	Dr Sukanya Pandey

Table 12: Michaelmas Term Timetable for Theory of Computation.

25 Models of Computation

This section is taught by Dr Barnaby Martin. [Check for recommended reading.](#)

25.1 Finite-State Automata and Regular Languages

25.2 Büchi Automata and Omega-Regular Languages

25.3 Linear-Time Logic and Translating Into Büchi Automata

26 Algorithms and Complexity

This section is taught by Dr Sukanya Pandey. The recommended reading for this topic is:

- Introduction to Algorithms (Cormen T.H., et al., 2009).

26.1 Asymptotic Complexity and Notation

26.2 Greedy Algorithms

26.3 Divide and Conquer Algorithms

26.4 Dynamic Programming Algorithms

26.5 Machinings in Graphs

Part VII

Appendix

27 Artificial Intelligence

27.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: Monday 15:30 - 16:30 in Term 1.

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00 in Term 2.

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: Thursday 14:30 - 15:30 in Term 2.

27.2 References

27.3 List of Figures

27.4 Reading List

27.4.1 Machine Learning

Deisenroth, M.P., Faisal, A.A. and Ong, C.S. (2020). Mathematics for Machine Learning. [online] Cambridge University Press. Available at: <https://mml-book.com/> [Accessed 3 Oct. 2026].

Hackeling, G. (2017). Mastering Machine Learning with scikit-learn. 2nd edn. Packt Publishing.

27.4.2 Ethics and Bias in AI

Various, will be added over time.

28 Data Science

28.1 Module Professors

Module Lead: Professor Thomas Erlebach

- Focus: Algorithms and networks.

- Email: thomas.erlebach@durham.ac.uk
- Office Hours: n/a.

Lecturer: Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00.

Lecturer: Doctor George Alex Koulieris

- Focus: Image processing and computer graphics.
- Email: georgios.a.koulieris@durham.ac.uk
- Office Hours: Please email to make an appointment.

28.2 References

28.3 List of Figures

28.4 Reading List

28.4.1 Machine Learning

Degroot, M.H. and Schervish, M.J. (2012). Probability and Statistics. Boston Mass.: Pearson.

29 Networks and Systems

29.1 Module Professors

Module Lead: Doctor Wanqing Tu

- Focus: Computer networks and wireless communications.
- Email: wanqing.tu@durham.ac.uk
- Office Hours: Thursday 11:00 - 12:00.

Lecturer: Doctor Thirupathaiah Vasantam

- Focus: Applied probability and performance modelling.
- Email: thirupathaiah.vasantam@durham.ac.uk
- Office Hours: Monday 14:00 - 15:00.

Lecturer: Doctor Carlton Shepherd

- Focus: Cyber security.
- Email: carlton.g.shepherd@durham.ac.uk
- Office Hours: Email to make an appointment.

29.2 References

29.3 List of Figures

29.4 Reading List

29.4.1 Databases

Connolly, T.M. and Begg, C.E. (2015). Database Systems: A practical approach to design, implementation, and management. 6th edn. Harlow, Essex, England: Pearson Education Limited.

29.4.2 Distributed Systems

Van Steen, M. and Tanenbaum, A.S. (2025). Distributed Systems. 4th edn. [online] Self-Published. Available at: <https://www.distributed-systems.net/index.php/books/ds4/> [Accessed 5 Oct. 2026].

30 Programming Paradigms

30.1 Module Professors

Module Lead: Doctor Stuart James

- Focus: Artificial intelligence, computer vision, human computer interaction, and digital humanities.
- Email: stuart.a.james@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Billy Moses Jr.

- Focus: Algorithms and biologically inspired computing.
- Email: william.k.moses-jr@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Mladen Ivkovic

- Focus: Performance prediction and optimisation.
- Email: mladen.ivkovic@durham.ac.uk
- Office Hours: Email to make an appointment.

30.2 References

30.3 List of Figures

31 Software Engineering

31.1 Module Professors

Module Lead: Doctor Craig Stewart

- Focus: Adaptive systems, HCI, and user modelling.
- Email: craig.d.stewart@durham.ac.uk

- Office Hours: Friday 11:00 - 12:00.

Lecturer: Doctor Nelly Bencomo

- Focus: Software engineering for autonomous, self-adaptive, AI and ML systems.
- Email: nelly.bencomo@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Gagangeet Aujla

- Focus: Data-driven and data-intensive cyber-physical systems and digital twins.
- Email: gagangeet.s.aujla@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Gokberk Kabacaoglu

- Focus: Machine learning and numerical applications.
- Email: gokberk.kabacaoglu@durham.ac.uk
- Office Hours: Thursday 14:00 - 15:00.

Lecturer: Doctor Ryan Crosby

- Focus: The assessment of novice programmers.
- Email: ryan.crosby@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Sukanya Pandey

- Focus: Graph theory.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Email to make an appointment.

31.2 References

31.3 List of Figures

32 Theory of Computation

32.1 Module Professors

Module Lead: Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.
- Email: barnaby.d.martin@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Sukanya Pandey

- Focus: Graph theory.
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- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Stefan Dantchev

- Focus: Combinatorics.
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- Office Hours: **To be confirmed.**

Lecturer: Doctor George Mertzios

- Focus: Algorithms and networks.
- Email: george.mertzios@durham.ac.uk
- Office Hours: **To be confirmed.**

32.2 References

32.3 List of Figures

32.4 Recommended Reading

32.4.1 Algorithms and Complexity

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C. (2009). Introduction to Algorithms. 3rd edn. MIT Press.